

EXAM 1

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Question 1

✓ 1 pt ↻ 1

Write the appropriate term for each definition below.

a) one possible result of an experiment ?

b) the probabilities of an event obtained from the cell at the intersection a row and a column Joint Probability

c) a coin for which the outcomes, heads and tails, are not equally likely
?

d) for any event A, the probability of A not happening is $P(\text{not } A) = 1 - P(A)$
?

e) the event that occurs when A doesn't occur ?

Question 2

✓ 1 pt ↻ 1

In a study, the data you collect is Letter grade (A,B,C,D,F).

This data is:

- ☐ Quantitative
- ☒ Qualitative

Question 3

✓ 1 pt ↻ 1

Determine whether the value 45% is a parameter or statistic:

A sample of men reveals 45% took calculus

- ☒ Statistic
- ☐ Parameter

Question 4

✓ 1 pt ↻ 1

Does this describe an observational study or an experiment?

The gender of children born in January were tallied

☐ Experiment

☒ Observational Study

● Question 5

✓ 1 pt ↻ 1

Determine whether the following studies would best be completed using a sample or population:

a. Determining the average weight of Labrador puppies.

Sample

b. Determining the average number of points per game for a famous basketball player.

Population

c. Determining the average grade of one statistics class.

Population

d. Determining the average number of eggs a salmon lays.

Sample

● Question 6

✓ 1 pt ↻ 1

Is this value from a discrete or continuous data set.

The average weight of newborn babies in ounces

☐ Discrete

☒ Continuous

● Question 7

✓ 1 pt ↻ 1

Choose the appropriate term for each definition below.

-Expected Value an approximation for the average value of a random variable over a large number of experiments

Probability dist. table the table that summarizes the possible values of a discrete random variable and the corresponding probabilities

Binomial Random Variable discrete random variable that represents the number of successes among n Bernoulli trials.

Probability histogram a graph that displays the possible values of a discrete random variable on the horizontal axis and the probabilities of those values on the vertical axis

Binomial Exp. an experiment in which there are exactly two possible outcomes - success and failure, with the probability of success $P(S)=p$ and the probability of failure $P(F)=1-p$.

● Question 8

✓ 1 pt ↺ 1

An online instructor wants to estimate the mean amount of time that students spend in the online homework system on a particular homework assignment. The instructor takes a random sample of 30 students and looks up the amount of time recorded by the system for that assignment.

Match the following:

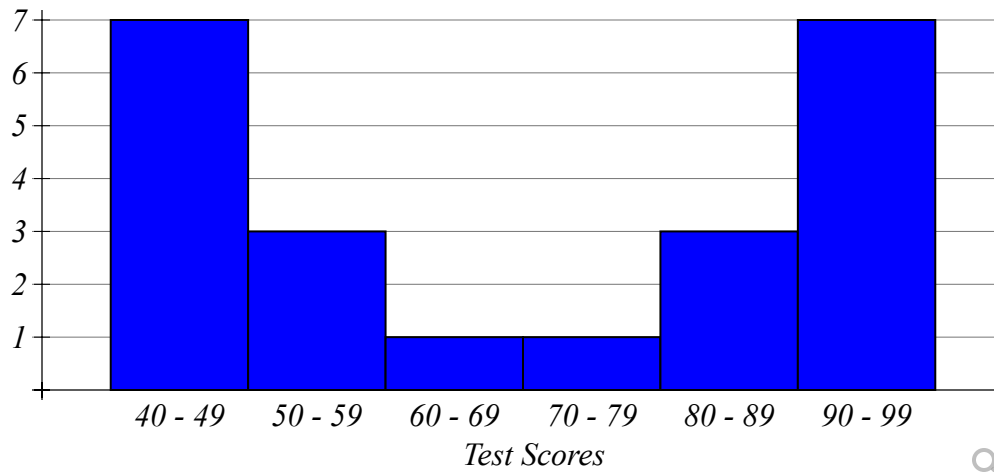
- ☒ e 30 students in that instructor's online class
- ☒ a mean time spent on a particular homework assignment for all students in that instructor's online class
- ☒ f a student in that instructor's online class
- ☒ d all students in that instructor's online class
- ☒ c time spent on a particular homework assignment
- ☒ b mean time spent on a particular homework assignment for 30 students in that instructor's online class

- a. parameter
- b. statistic
- c. variable
- d. population
- e. sample
- f. individual

● Question 9

✓ 1 pt ↺ 1

Which distribution type best fits the following data?



Choose the type of distribution:



● Question 10

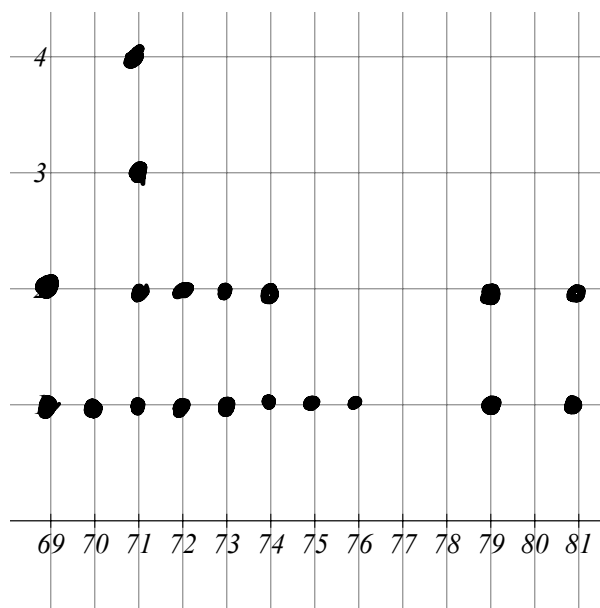
✓ 1 pt ↺ 1

The following data set is given:

79	76	71	81	69
73	71	70	79	73
74	71	72	81	74
72	75	71	69	

~~69~~ - 2 ~~76~~ - 1
~~70~~ - 1 ~~77~~ - 0
~~71~~ - 4 ~~78~~ - 0
~~72~~ - 2 ~~79~~ - 2
~~73~~ - 2 ~~80~~ - 0
~~74~~ - 2 ~~81~~ - 2
~~75~~ - 1

Construct a dot plot:



Question 11

✓ 1 pt ↻ 1

Here is a data set summarized as a stem-and-leaf plot:

Stem	Leaf
5	0 0 0 1 3 4 4 7 7 7 7 = 11
6	0 0 0 1 1 1 2 2 4 4 5 6 6 7 7 8 = 17
7	0 3 6 6 9 = 5
8	2 5 5 = 3

How many data values are in this data set? 36 ~~x~~ ← This is 35

What is the minimum value in the last class? 82

What is the minimum value in the entire sample? 50

How many of the original values are greater than 70? 7

Question 12

✓ 1 pt ↻ 1

How many outcomes would there be in the sample space for rolling 5 dice and flipping 3 coins?

Question 13

✓ 1 pt ↻ 1

Dice

$$6 \text{ sides} \rightarrow 6^5 = 7,776$$

Coin

$$2 \text{ sides} \rightarrow 2^3 = 8$$

outcomes

$$7,776 \times 8 = 62,208$$

Occasionally an airline will lose a bag. A small airline has found it loses an average of 2.8 bags each day. Find the probability that, on a given day,

(a) The airline loses exactly five bags.

0.0872

(b) The airline loses fewer than six bags.

0.9349

(c) The airline loses more than three bags.

0.308

X

Round all answers to four decimal places.

$$P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$\lambda = 2.8$$

$$e = 2.71828 \leftarrow e^{-2.8}$$

$$a) k=5 \quad P(X=5) = \frac{e^{-2.8} \cdot 2.8^5}{5!} = \frac{0.06081 (172.1037)}{120} = 0.0872$$

$$b) P(X < 6)$$

$$P(0) = \frac{e^{-2.8} \cdot 2.8^0}{0!} = 0.0608$$

$$P(1) = \frac{e^{-2.8} \cdot 2.8^1}{1!} = 0.1703$$

$$c) P(X > 3)$$

$$P(X > 3) = 1 - P(X \leq 3)$$

$$0.0608 + 0.1703 + 0.2384 + 0.2225 = 0.6920 \quad \checkmark 1 \text{ pt } \odot 1$$

Question 14

$$P(2) = \frac{e^{-2.8} \cdot 2.8^2}{2!} = 0.2384$$

$$P(3) = \frac{e^{-2.8} \cdot 2.8^3}{3!} = 0.2225$$

$$P(4) = \frac{e^{-2.8} \cdot 2.8^4}{4!} = 0.1667$$

$$P(5) = \frac{e^{-2.8} \cdot 2.8^5}{5!} = 0.0872$$

$$0.9349$$

$$1 - 0.6920 = 0.3080$$

Carolyn sampled the ages from the list of the Iowa's millionaires and observed the following data:

30	40	83	66	83
59	23	35	65	83
68	68	41	69	34
46	54	78	66	94
68	63	56	47	41

$20 \leq x < 30 \rightarrow 1$
 $30 \leq x < 40 \rightarrow 3$
 $40 \leq x < 50 \rightarrow 5$
 $50 \leq x < 60 \rightarrow 3$
 $60 \leq x < 70 \rightarrow 8$
 $70 \leq x < 80 \rightarrow 1$
 $80 \leq x < 90 \rightarrow 3$

$90 \leq x < 100 \rightarrow 1$

Use a grouping with the first class' lower limit 20 and the class width of 10 to:

- Create the frequency table:

Ages	Freq
$20 \leq x < 30$	1
$30 \leq x < 40$	3
$40 \leq x < 50$	5
$50 \leq x < 60$	3
$60 \leq x < 70$	8
$70 \leq x < 80$	1
$80 \leq x < 90$	3
$90 \leq x < 100$	1

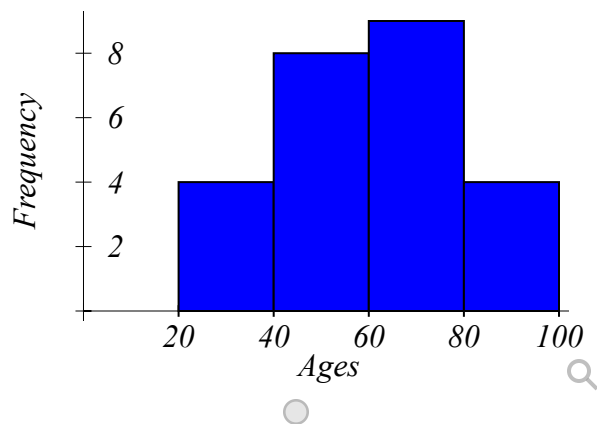
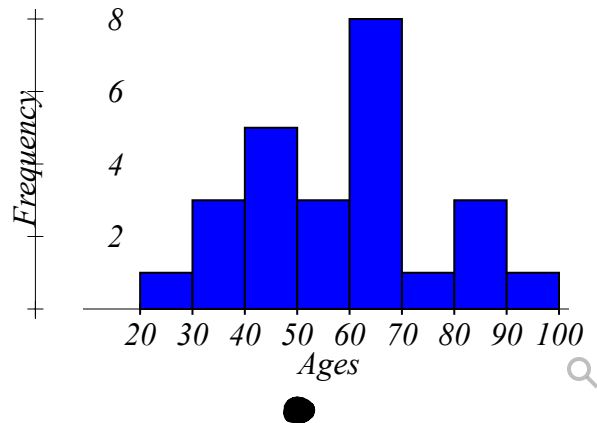
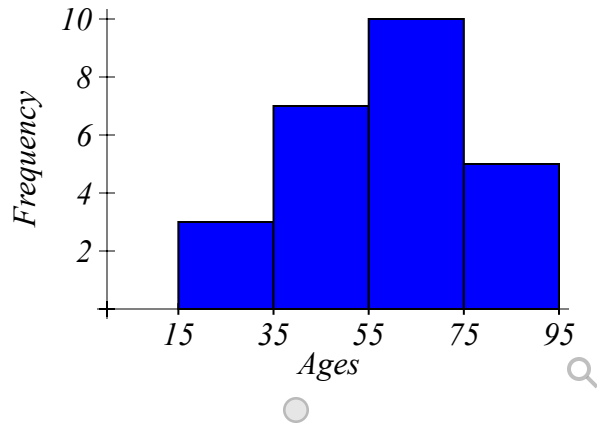
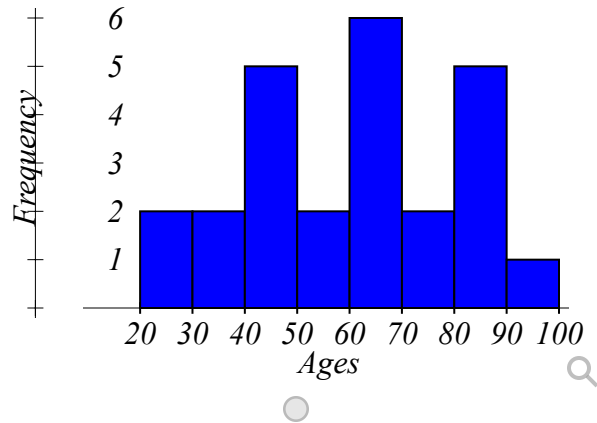
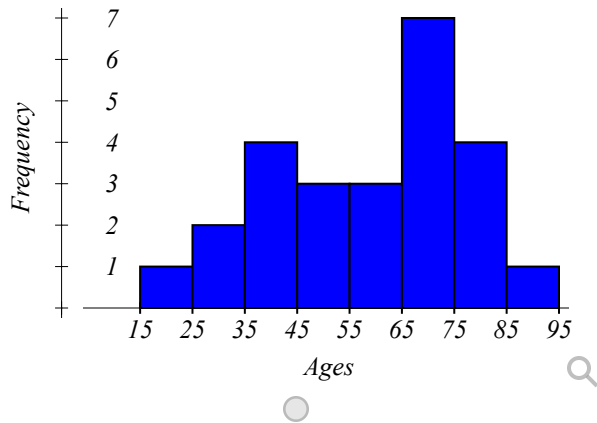
Ages	Freq
$15 \leq x < 35$	3
$35 \leq x < 55$	7
$55 \leq x < 75$	10
$75 \leq x < 95$	5

Ages	Freq
$20 \leq x < 40$	4
$40 \leq x < 60$	8
$60 \leq x < 80$	9
$80 \leq x < 100$	4

Ages	Freq
$15 \leq x < 25$	1
$25 \leq x < 35$	2
$35 \leq x < 45$	4
$45 \leq x < 55$	3
$55 \leq x < 65$	3
$65 \leq x < 75$	7
$75 \leq x < 85$	4
$85 \leq x < 95$	1

Ages	Freq
$20 \leq x < 30$	2
$30 \leq x < 40$	2
$40 \leq x < 50$	5
$50 \leq x < 60$	2
$60 \leq x < 70$	6
$70 \leq x < 80$	2
$80 \leq x < 90$	5
$90 \leq x < 100$	1

- Create the frequency histogram:



Identify the shape of the distribution of the data:

Skewed left

● Question 15

✓ 1 pt ↺ 1

Consider a small shop with only one checkout counter. The arrivals of customers are distributed throughout the day such that a line rarely forms at the counter. Suppose the arrival times of customers at the counter follows the Exponential Distribution: $X \sim \text{Exp}(0.4)$.

a. What is the mean time (in minutes) between customers?

$$\mu = \underline{2.5} \quad \leftarrow \mu = \frac{1}{\lambda} = \frac{1}{0.4} = 2.5$$

b. Determine the following probability.

$$P(2 < X < 10) = \underline{0.4310} \quad \leftarrow P(X \leq x) = 1 - e^{-\lambda x}$$

$$\begin{aligned} P(2 < X < 10) &= F(10) - F(2) \\ &= (1 - e^{-0.4(10)}) - (1 - e^{-0.4(2)}) \\ &= e^{-0.8} - e^{-4.0} \end{aligned}$$

(Include four decimal places)

$$= 0.449329 - 0.018316 = 0.4310$$